

Week 3

Summarizing Distributions

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Speaker notes

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Speaker notes

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The Importance of Summarizing Distributions

Speaker notes

- probability distributions are really flexible objects. that's a great strength because they can represent a lot of different things
- The number of friends on a social network.

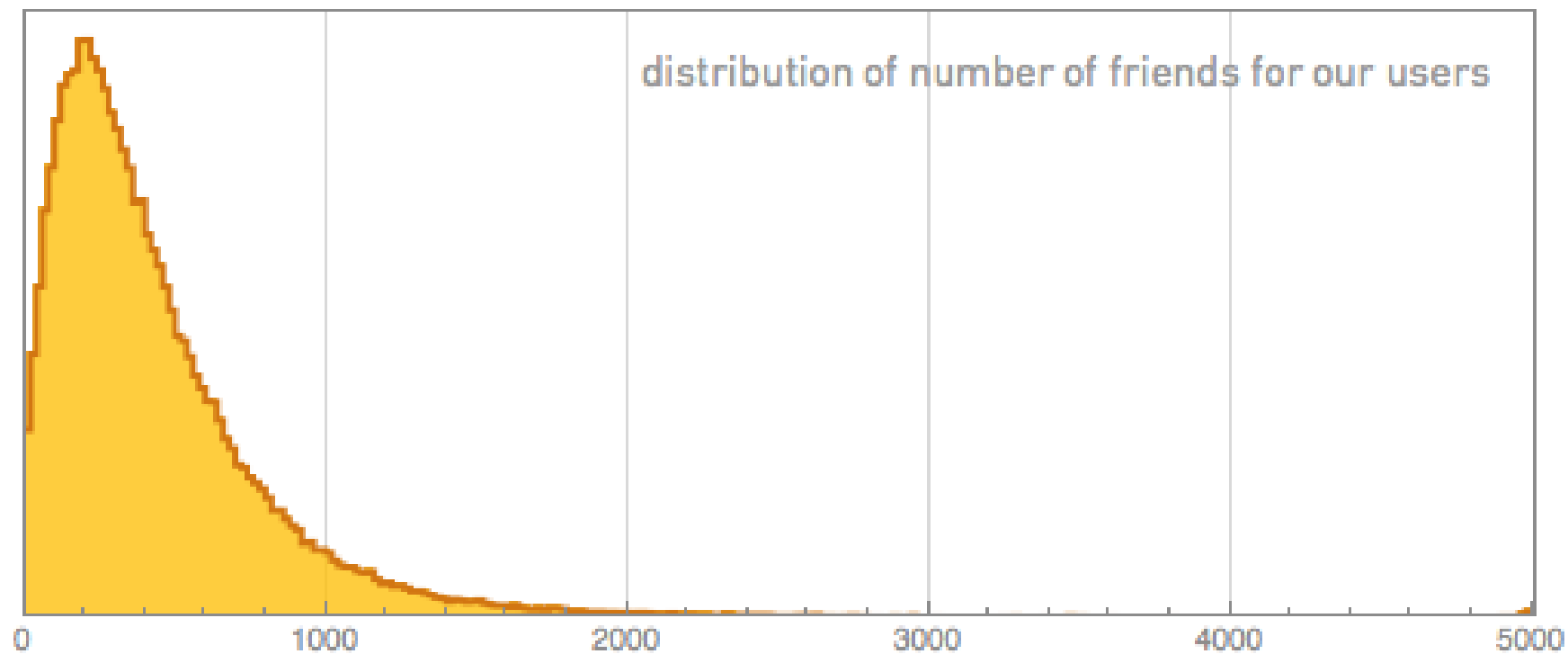
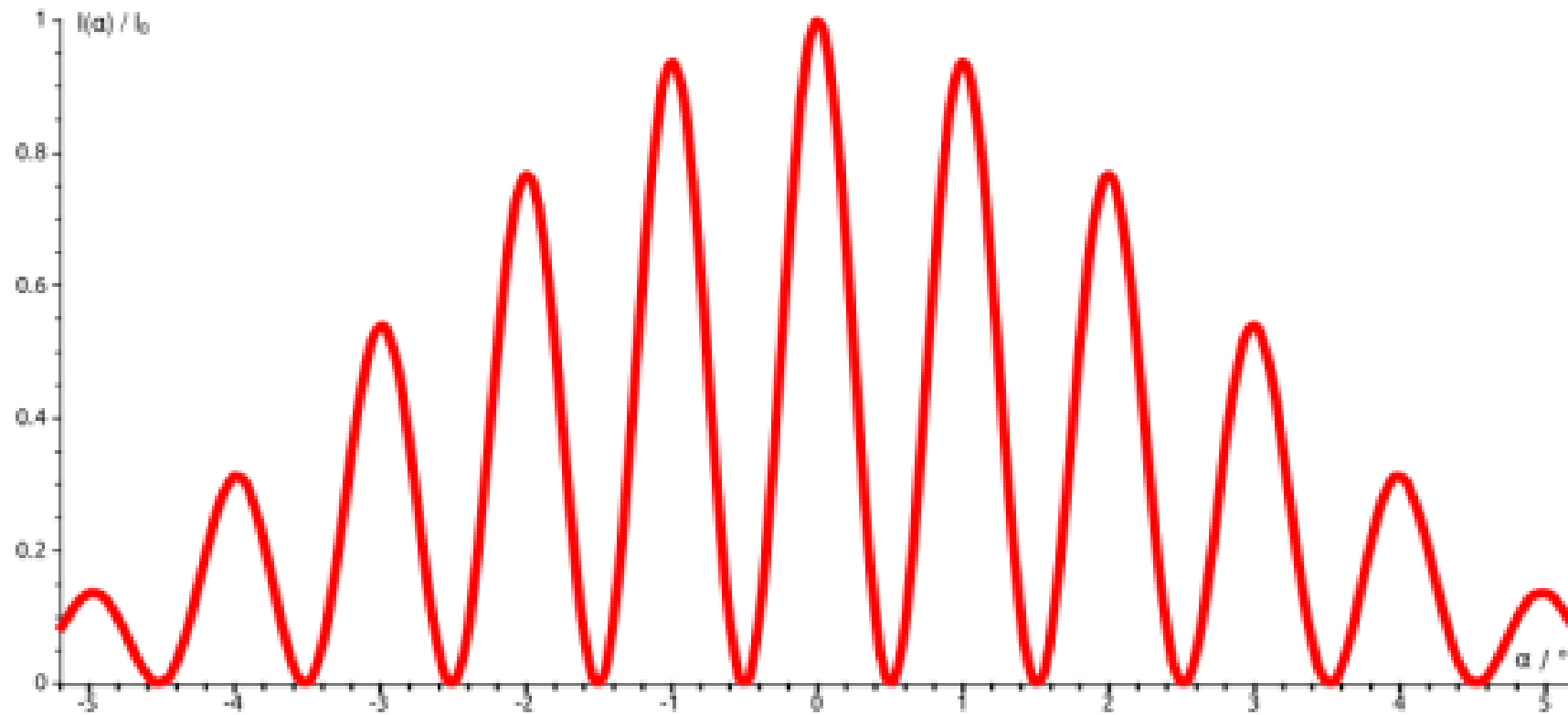
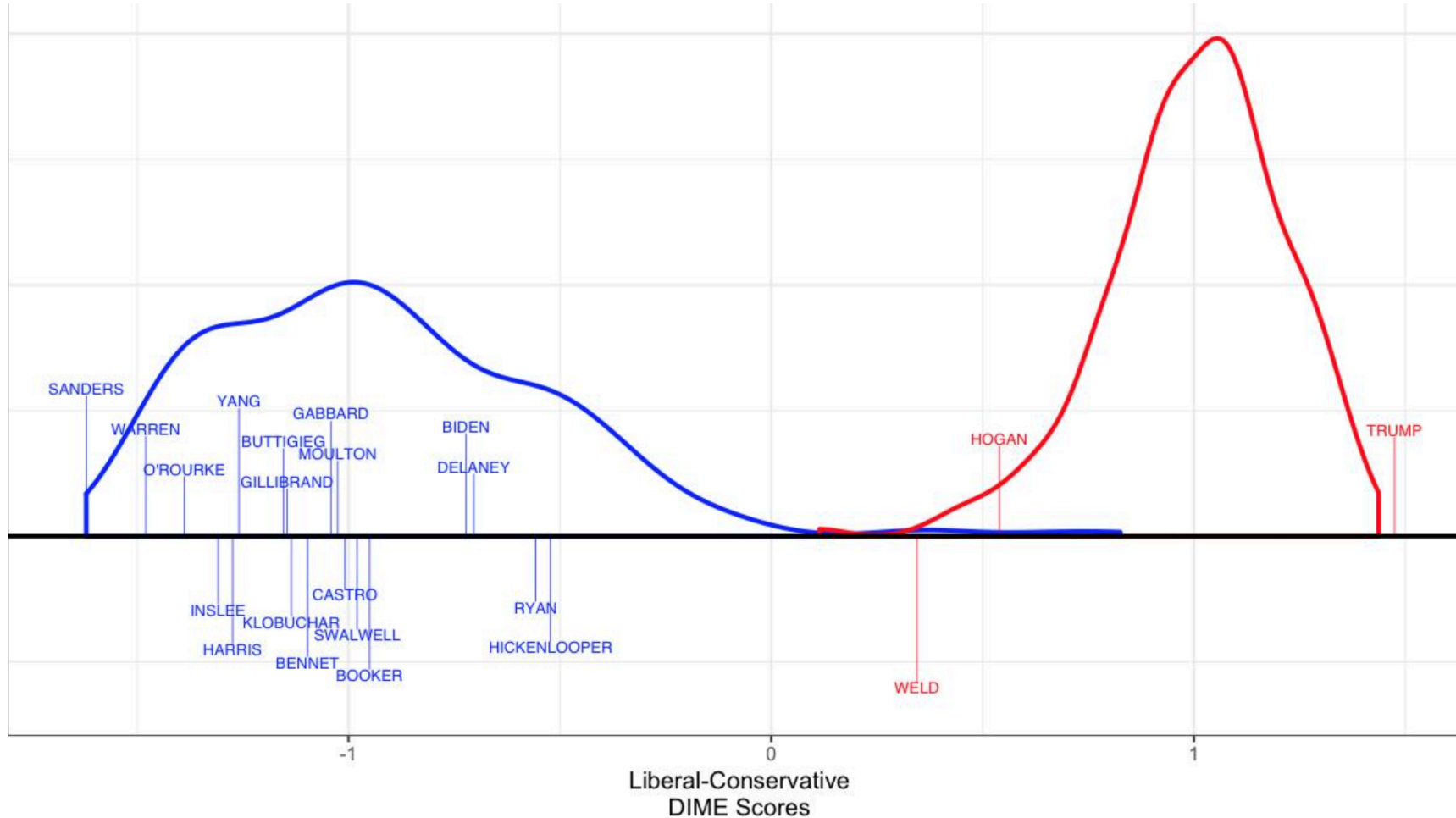


Image: stephenwolfram.com. Data Science of the Facebook World.

Speaker notes

- The location of photons in a physics experiment.





Source: Database on Ideology, Money in Politics, and Elections (DIME).

Speaker notes

- The liberal or conservative lean of members of congress.
- All of these example only use one random variable, and their distributions have very different shapes.
- To represent all of these, we need a flexible mathematical object. We can change a distribution in an infinite number of ways. move it up a bit in one place, move it down in another. That's what we need to represent the rich variety of phenomena that we want to model.
- But there's a down side to all this flexibility...

The Downside of Flexibility

Highly flexible objects present obstacles.

- Cannot specify density for every value
 - Cannot communicate or reason about every value
 - Cannot estimate density at every value given data
- Why not restrict to distributions in a parametric family?
- The real world isn't that neat!
 - “Agnostic” approach: minimal assumptions on the model

Why Summarize?

_ Summary tools are the hooks we use to estimate, reason about, and communicate about random variables._

Foundational Summary Tools

One Random Variable:

- Expectation: represents the center of a distribution
- Variance: represents the dispersion of a distribution

Two Random Variables

- Covariance: a generalization of variance
- Correlation: an intuitive measure of linear association

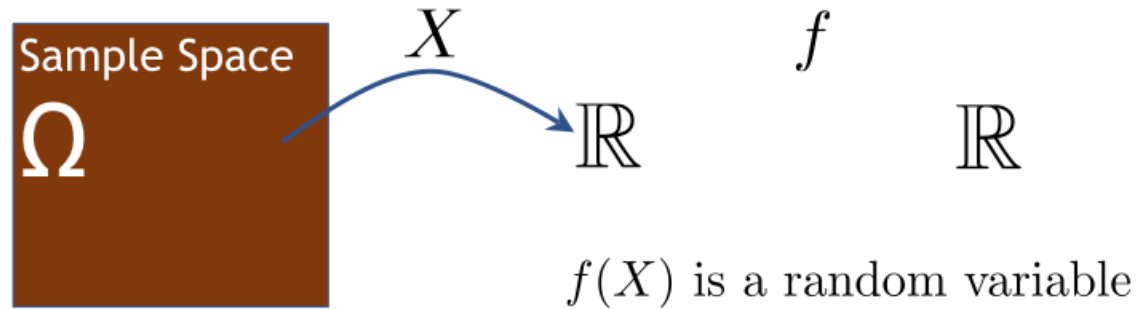
Speaker notes

- In this unit, we're going to cover a few of the most foundational summary tools out there.
- We'll start with a single random variable.
- Expectation, as we'll see, is like a mean, but a version of the mean for random variables. Variance is about how spread out a distribution is.
- Then we'll move on to pairs of random variables - how do we summarize relationships?
- covariance and correlation are tools that capture the strength of a linear relationship in a single number
- At the end of this unit, you'll be much better placed to study, and communicate about random variables.

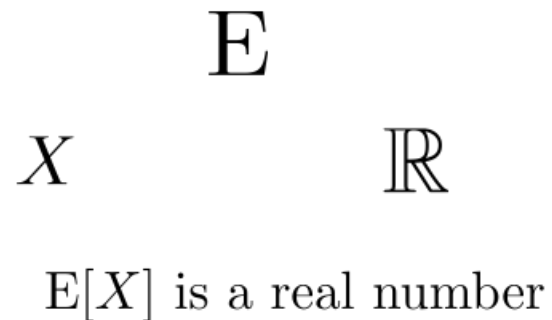
Review of Operators and Functions

Review of Operators and Functions

Function $f(x) = x^2$



Operator E



Speaker notes

- As you start reading about expectation, it's good to be reminded that expectation is an example of an operator. Actually, all the summary tools we're going to cover are operators, so let's quick review what that word means. What's the difference between a function and an operator.
- When we say a function of a random variable, we mean a function from $\mathbb{R}$ to $\mathbb{R}$
- A function takes the output of a random variable as input and maps it to a new real number
- putting a random variable into a function gives you another random variable
- Expectation is an operator, which means it takes an entire random variable as input
- So, for example, E

Reading: Summarizing Distributions

Reading Assignment

Read *Foundations of Agnostic Statistics* , page 44 to the top of page 47, stopping after you read theorem 2.1.5.

Lightboard: Expectation of a Discrete Random Variable

Expectation of a Discrete Random Variable

Expectation for Discrete Random Variables

$$\mathbb{E}[X] = \sum_{x \in \text{supp}[X]} x f(x) \quad \text{Ex 1}$$

$X \sim \text{Bernoulli}(\alpha)$ $f(x) =$

$$\begin{cases} \alpha, & x = 1 \\ 1 - \alpha, & x = 0 \\ 0, & \text{otherwise} \end{cases}$$

\$

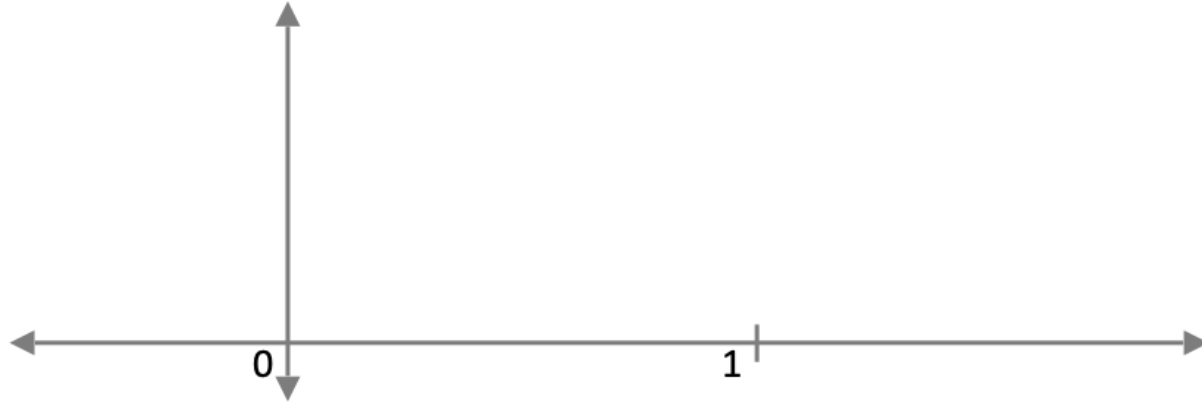
$$\mathbb{E}[X] = 0 \cdot (1 - \alpha) + 1 \cdot \alpha = \alpha$$

Ex 2: $Y \sim \text{Geometric}(\beta)$
 Let J_i be the event printer jams
 day i . $P(J_i) = \beta$, all ind.

Let Y be days until first jam $P(Y=1) = \beta$
 $P(J_1) = \beta$ $P(Y=2) = P(J_1^c, J_2) = (1-\beta)\beta$
 $P(Y=3) = (1-\beta)^2\beta$

Lightboard: Expectation of a Continuous Random Variable

Expectation of a Continuous Random Variable



Expectation for Continuous Random Variables

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} xf(x)dx \quad \text{Ex 1:}$$

$$X \sim \text{Uniform}(a, b) \quad f_X(x) =$$

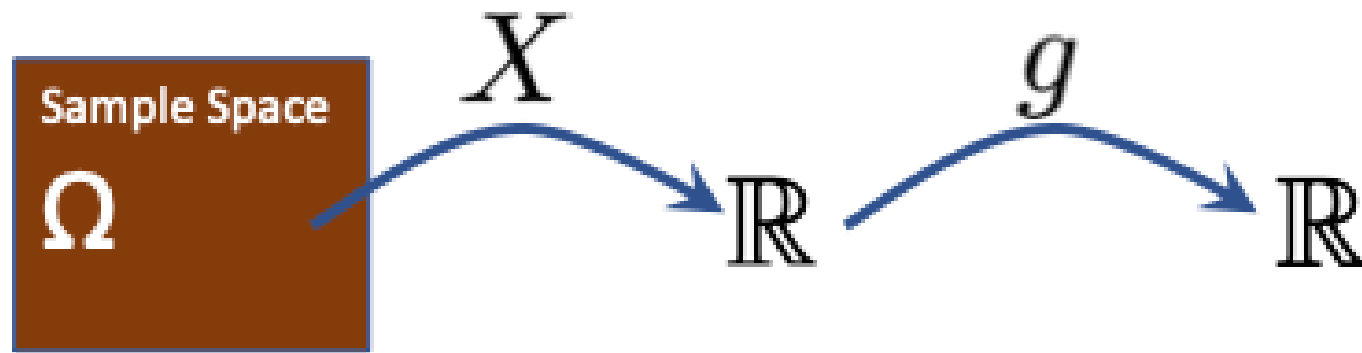
$$\begin{cases} \frac{1}{b-a}, & a < x < b \\ 0, & \text{otherwise} \end{cases}$$

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} xf(x)dx$$

$$= \int_a^b x \cdot \frac{1}{b-a} dx + \int_{-\infty}^a 0 dx + \int_b^{\infty} 0 dx$$

Law of the Unthinking Statistician

Law of the Unthinking Statistician (LOTUS)



Law of the Unthinking Statistician (LOTUS)

X is a DRV with pmf f
 $g(X)$ is a DRV with pmf $f'(y) =$
 $P(g(X) = y) = \sum_{\{x \in \mathcal{R} : g(x) = y\}} f(x)$

want: $E[g(X)]$ \note[item]{We want the expectation of $g(X)$. we could first compute the pmf of $g(X)$ then find the expectation.

That can be hard. \note[item]{Instead, we have this really useful formula} \note[item]{This is called LOTUS – I think because it's so easy, that once you level up enough as a

Proof sketch:
$$E[g(X)] = \sum_y y f'(y)$$

$$= \sum_y \sum_{\{x \in \mathcal{R} : g(x) = y\}} y f(x)$$

$$= \sum_y \sum_{\{x \in \mathcal{R} : g(x) = y\}} y f(x)$$

Reading: Linearity of Expectation

Reading Assignment

Read *Foundations of Agnostic Statistics* , pages 47–50, stopping at 2.1.2.

Lightboard: Linearity of Expectation

Linearity of Expectation

Assume RV C with pmf $f(x) =$
 $\begin{cases} 1, & \text{if } x=c \\ 0, & \text{otherwise} \end{cases}$
 then C is a constant. $E[C] =$
 $\sum_x x f(x) = c f(c) = c$
 *usually write c for RV.

Let X, Y be DRV's. $a \in \mathbb{R}$
 {All of these are true
 for discrete and continuous RVs,
 but we'll prove with discrete

$$\begin{aligned} E[X + Y] &= \sum_{x \in X(\Omega), y \in Y(\Omega)} (x + y) f(x, y) \\ &= \sum_x \sum_y x f(x, y) + \sum_y \sum_x y f(x, y) \\ &= \sum_x x \sum_y f(x, y) + \sum_y y \sum_x f(x, y) \\ &= \sum_x x f_X(x) + \sum_y y f_Y(y) \\ &= E[X] + E[Y] \end{aligned}$$

Speaker notes

- We introduced expectation by comparing it to the mean, and yes, it is a mean, but that makes it sound like just a fixed number
- It's important to start thinking about expectation as an operator. from the space of random variables to $\mathbb{R}$
- As an operator, expectation has important properties. Let's quickly see a few of them

Reading: Moments and Variance

Reading Assignment

Read *Foundations of Agnostic Statistics* , section 2.1.2.

Lightboard: Moments and Variance

Moments and Variance

(raw) moments:

$$\mathbb{V}[aX] = \mathbb{E}[(aX)^2] - \mathbb{E}[aX]^2 = a^2 \mathbb{V}[X]$$

$$\mathbb{V}[X + c] = \mathbb{E}[(X + c - \mathbb{E}[X + c])^2]$$

1. $\mathbb{E}[X]$ 2. $\mathbb{E}[X^2]$
 3. $\mathbb{E}[X^3]$...
 What do the moments measure? Well, one way to think about them is that as the power gets higher, the larger values of X dominate the smaller ones, so it's like you're checking for how much probability is in the tails.
 Possibly omit moments discussion? we will refer to them in method of moments, but can always review
 Define variance. $\mathbb{V}[aX] = \mathbb{E}[\text{big}[\dots]]$

Reading: Mean Squared Error (MSE)

Reading Assignment

Read *Foundations of Agnostic Statistics* , section 2.1.3.

Lightboard: Mean Squared Error (MSE) - Lightboard

The Expected Value and MSE

Proof that $\mathbb{E}[X]$ minimizes MSE

$$\begin{aligned}
 \text{MSE} &= \mathbb{E}[(X - c)^2] \quad \text{variable } c \\
 \mathbb{E}[(X - c)(X - c)] &= \mathbb{E}[X^2 - 2cX + c^2] \\
 &= \mathbb{E}[X^2] - 2c\mathbb{E}[X] + c^2 \\
 &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 + \mathbb{E}[X]^2 - 2c\mathbb{E}[X] + c^2 \\
 \text{MSE} &= \text{Var}[X] + (\mathbb{E}[X] - c)^2
 \end{aligned}$$

$$\begin{aligned}
 \text{min}_{c} \text{MSE} & \text{ as a function of } c \\
 \text{MSE} &= \mathbb{E}[(X - c)^2] \\
 \mathbb{E}[X^2 - 2cX + c^2] \\
 \frac{\partial \text{MSE}}{\partial c} &= \mathbb{E}[X^2] - 2c\mathbb{E}[X] + c^2 \\
 2\mathbb{E}[X] - 2c &= 0 \\
 \mathbb{E}[X] - c &= 0 \quad \boxed{c = \mathbb{E}[X]}
 \end{aligned}$$

Covariance and Correlation

Measuring Linear Dependency

Understanding Relationships

Most of the really important questions in data science are about *relationships* .

- Bitterness of coffee
- Roasting temperature and time

Speaker notes

- If I knew everything about the marginal distribution of bitterness by itself, maybe I could publish a blog post.
- But, if I can understand how several variables relate to one another, I can do more useful things than write a blog-post – I can change my world and drink better coffee. I could (a) Generate hypotheses; (b) Make predictions of bitterness for different roast profiles; (c) Maybe build a new coffee theory; or (d) Drink better coffee.

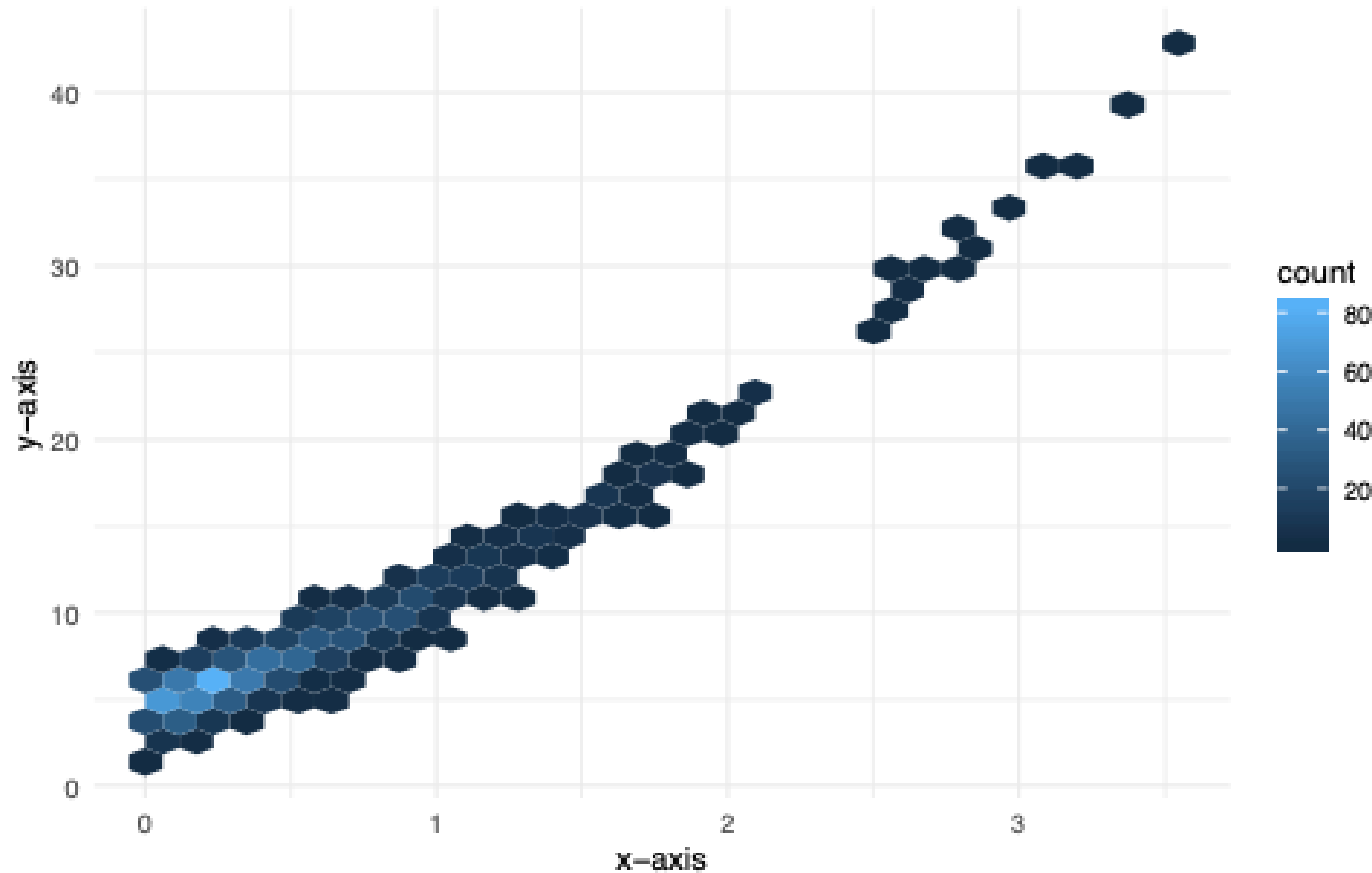
standout

How can we make sense of the wide variety of relationships among variables?

Speaker notes

No notes on this slide.

The Joint Distribution



Speaker notes

- One answer is the joint distribution. This fully describes the relationship between random variables
- But: (1) It's a lot of information; (2) It is information that doesn't come as a part of our starter-pack – we are not provided a manual that has the joint distribution in it;
 3. Sometimes, even
- We really need tools that are easier to work with

Measuring Dependency

A good first question about two random variables:

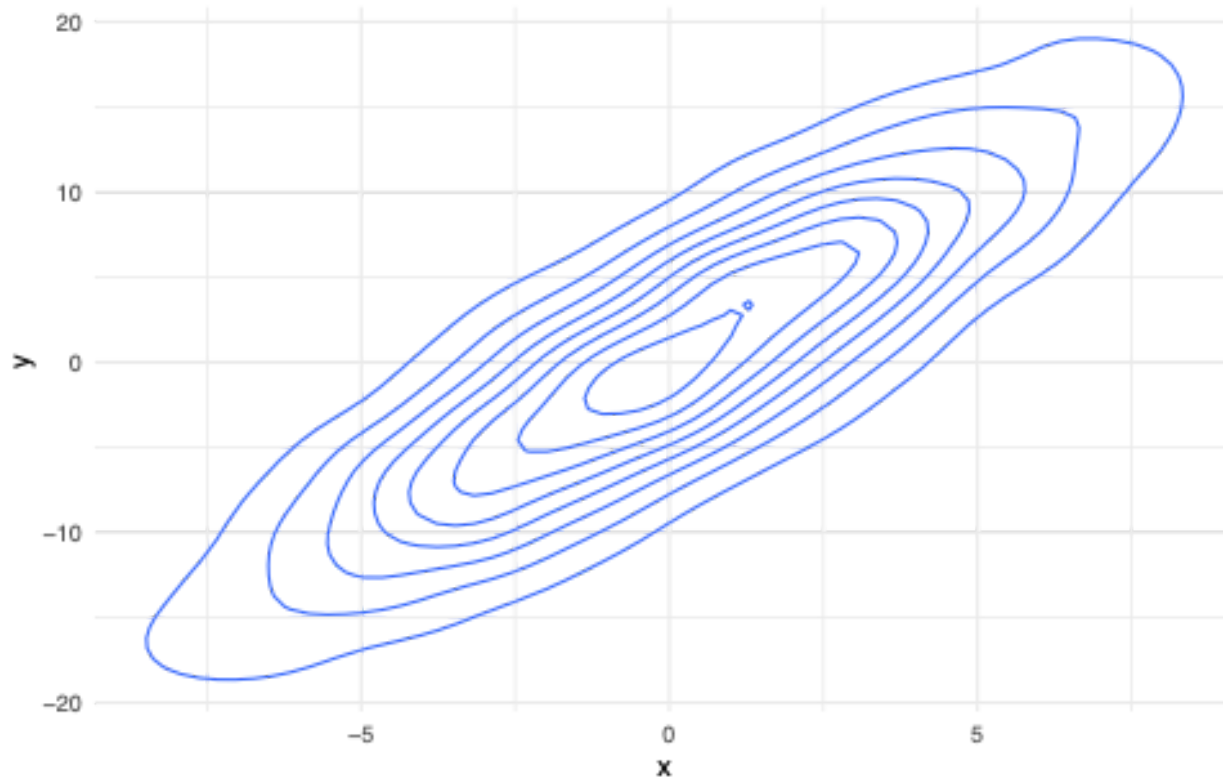
How strong is the relationship?

Two common tools to answer this question:

- Covariance
- Correlation

Intuition for Covariance

$$\text{cov}(X, Y) = \mathbb{E}[(X - E[X])(Y - E[Y])]$$



Speaker notes

- Let's take a quick look at the covariance formula. Just like in variance, we're looking at deviations away from a mean. But here, we're multiplying X's deviations, by Y's deviations.
- Let me mark a point here, that's a high probability point. Notice that X's deviations and Y's are positive, so when we multiply them, we get something positive.
- What about on the other side? these are also high probability points. notice that both deviations are negative, so the product is again positive.
- This is what we mean by two random variables varying together. they will have positive covariance

Reading Assignment

Reading Assignment

Note: This is a reading call, we're just placing it here for **organization**. Read *Foundations of Agnostic Statistics* , pages 59–62, stopping at Correlation.

Variance of Sums

Variance of Sums

Insert content from old section of course: 4.10 Variance of Sums

Alternative Formula for Covariance

Alternative Formula for Covariance

$$\text{cov}[X, Y] = \text{E}[(X - \text{E}[X])(Y - \text{E}[Y])]$$

Learnosity

Speaker notes

No notes on this slide.

Learnosity: $V[X - Y]$

You have just read theorem 2.2.3, which says, in part:

Variance of sums

$$V[X + Y] = V[X] + 2\text{cov}[X, Y] + V[Y]$$

1. If X and Y are independent — that is, $\text{cov}[X, Y] = 0$ — which is larger? (a) $V[X - Y]$; (b) $V[X + Y]$; (c) They are the same. (d) We don't have enough information.
2. If X and Y are not independent — that is, $\text{cov}[X, Y] \neq 0$ — which is larger? (a) $V[X - Y]$; (b) $V[X + Y]$; c. They are the same. (d) We don't have enough information.

Learnosity: $V[X - Y]$ (cont.)

1. Under what circumstances would it be possible for $V[X]$ and $V[Y]$ to “cancel out”?
2. Notice that $V[X + (-1 * X)] = V[X - X] = V[0] = 0$. Use this to find a simple formula for $cov[X, X]$.

Properties of Covariance Lightboard

Properties of Covariance

For random variables X, Y, Z, W and $a, b \in \mathbb{R}$,

Lightboard: Linearity of Covariance

Note: This is Lightboard. We're just placing it here for organization. For random variables X, Y, Z, W and constants a, b ,

$$\text{cov}[aX, bY] = E[aXbY] - E[aXE[BY]] = abE[XY] - abE[X]E[Y]$$

$$\begin{aligned}\text{cov}[X + Y, Z] &= E[(X + Y) \cdot Z] - E[X + Y]E[Z] \\ &= E[XZ + YZ] - (E[X] + E[Y])E[Z] = E[XZ] + E[YZ] - E[X]E[Z] \\ &= \text{cov}[X, Z] + \text{cov}[Y, Z]\end{aligned}$$

$$\text{b.s.a } \text{cov}[X, Y + Z] = \text{cov}[X, Y] + \text{cov}[X, Z]$$

$$\begin{aligned}\text{cov}[X + Y, Z + W] &= \text{cov}[X, Z + W] + \text{cov}[Y, Z + W] \\ &= \text{cov}[X, Z] + \text{cov}[X, W] + \text{cov}[Y, Z] + \text{cov}[Y, W]\end{aligned}$$

Correlation

Correlation is Rescaled Covariance

Definition 2.2.5

Correlation is a rescaled derivative of covariance that captures the _linear dependence_ between two random variables.

$$\rho[X, Y] = \frac{\text{cov}[X, Y]}{\sigma[X]\sigma[Y]}$$

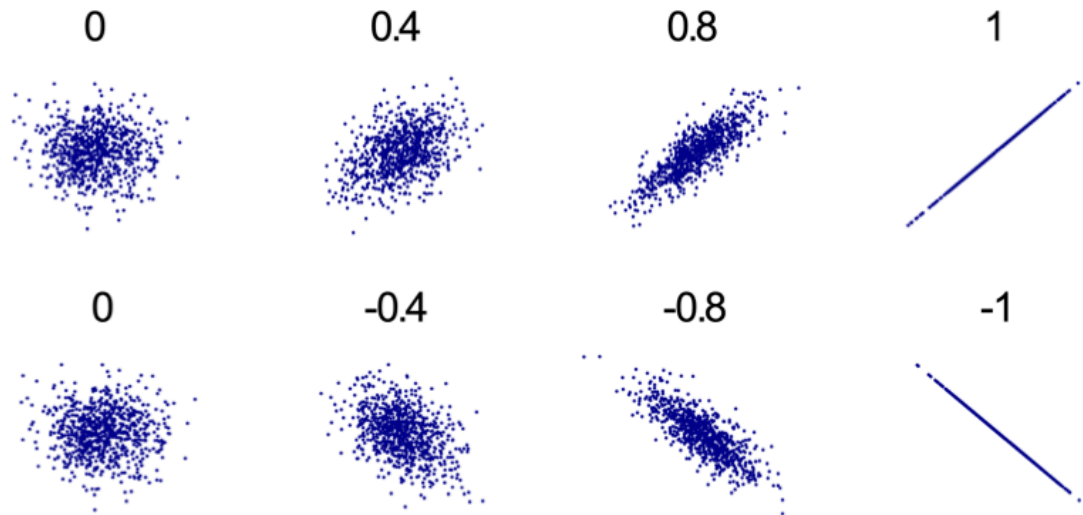
Society has a plain-language usage of *correlation* that is probably closer in usage to *covariance* than to *correlation* .

Speaker notes

- Schooling is correlated with earnings.
- In general it is a stand-in for “Is associated” with.
- But this isn’t precisely correct.

Correlation Measures Linear Dependency

Correlation for example distributions:



Source: en.wikipedia.org/wiki/Correlation_and_dependence

Speaker notes

- The presence of a correlation implies that there is some linear dependence between two random variables.
-
-

Not All Dependency is Linear

Example distributions with zero correlation:



Source: en.wikipedia.org/wiki/Correlation

and dependence

Speaker notes

- But, the lack of a correlation (and the lack of a linear dependence) does not mean that there is no dependence – does not mean that the random variables are independent.

-

Reading: Correlation

- Feel free to skip footnote 12. The authors are showing off.

Reading Assignment

Read pages 62-65, stopping before you get to example 2.2.9.

Theorem 2.2.7

The third and fourth bullet points can be more clearly stated.

– If a and b are either both positive or both negative, $\rho[aX + c, bY + d] = \rho[X, Y]$

- If a and b have opposite signs, $\rho[aX + c, bY + d] = -\rho[X, Y]$

Covariance, Correlation, and Independence

Independent Variables Have Zero Correlation

i Independence $\Rightarrow \rho = 0$

If X and Y are independent random variables, then:
– $E[XY] = E[X] E[Y]$

- $\text{cov}[X, Y] = 0$
- $\rho[X, Y] = 0$
- $V[X + Y] = V[X - Y] = V[X] + V[Y]$

Speaker notes

- The really neat curiosity, that we showed earlier is that covariance or correlation equals zero does not imply that the variables are independent.
- As our understanding of relationships permits non-linear dependencies, we're going to need a more flexible model.
Welcome in machine learning!

Independent Variables Have Zero Correlation

Speaker notes

No notes on this slide.

Wrap-Up of Correlation

Wrap-Up of Correlation

Note: This is a solo-lecture. We are just placing this here for organization.

Coding Activity

Note: This will be a README. We're just placing it here for organization.

Speaker notes

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